

A VCG-Based Mechanism for Electronic Commerce Applications

SUBMISSION 42

In auction theory, a Vickrey–Clarke–Groves (VCG) auction is a type of sealed-bid auction of multiple items. Bidders submit bids that report their valuations for the items, without knowing the bids of the other people in the auction. The auction system assigns the items in a socially optimal manner: it charges each individual the harm they cause to other bidders. It also gives bidders an incentive to bid their true valuations, by ensuring that the optimal strategy for each bidder is to bid their true valuations of the items. (Thanks for this text, Wikipedia.) This paper shows that the same idea works for electronic commerce applications.

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1 Introduction

Since we're all intimately familiar with VCG, let's learn about Wireless Sensor Networks by using the standard ACM demo article text from here on.

As a new technology, Wireless Sensor Networks (WSNs) has a wide range of applications [???], including environment monitoring, smart buildings, medical care, industrial and military applications. Among them, a recent trend is to develop commercial sensor networks that require pervasive sensing of both environment and human beings, for example, assisted living [???] and smart homes [???].

“For these applications, sensor devices are incorporated into human cloths [????] for monitoring health related information like EKG readings, fall detection, and voice recognition”.

While collecting all these multimedia information [?] requires a high network throughput, off-the-shelf sensor devices only provide very limited bandwidth in a single channel: 19.2 Kbps in MICA2 [?] and 250 Kbps in MICAz.

In this article, we propose MMSN, abbreviation for Multifrequency Media access control for wireless Sensor Networks. The main contributions of this work can be summarized as follows.

- To the best of our knowledge, the MMSN protocol is the first multifrequency MAC protocol especially designed for WSNs, in which each device is equipped with a single radio transceiver and the MAC layer packet size is very small.
- Instead of using pairwise RTS/CTS frequency negotiation [????], we propose lightweight frequency assignments, which are good choices for many deployed comparatively static WSNs.
- We develop new toggle transmission and snooping techniques to enable a single radio transceiver in a sensor device to achieve scalable performance, avoiding the nonscalable “one control channel + multiple data channels” design [?].

2 MMSN Protocol

2.1 Frequency Assignment

We propose a suboptimal distribution to be used by each node, which is easy to compute and does not depend on the number of competing nodes. A natural candidate is an increasing geometric sequence, in which

$$P(t) = \frac{b^{\frac{t+1}{T+1}} - b^{\frac{t}{T+1}}}{b - 1}, \quad (1)$$

where $t = 0, \dots, T$, and b is a number greater than 1.

In our algorithm, we use the suboptimal approach for simplicity and generality. We need to make the distribution of the selected back-off time slice at each node conform to what is shown in Equation (1). It is implemented as follows: First, a random variable α with a uniform distribution within the interval $(0, 1)$ is generated on each node, then time slice i is selected according to the following equation:

$$i = \lfloor (T + 1) \log_b [\alpha(b - 1) + 1] \rfloor.$$

It can be easily proven that the distribution of i conforms to Equation (1).

So protocols [??????] that use RTS/CTS controls¹ for frequency negotiation and reservation are not suitable for WSN applications, even though they exhibit good performance in general wireless ad hoc networks.

¹RTS/CTS controls are required to be implemented by 802.11-compliant devices. They can be used as an optional mechanism to avoid Hidden Terminal Problems in the 802.11 standard and protocols based on those similar to ? and ?.

ALGORITHM 1: Frequency Number Computation

Input: Node α 's ID (ID_α), and node α 's neighbors' IDs within two communication hops.

Output: The frequency number ($FreNum_\alpha$) node α gets assigned.

$index = 0; FreNum_\alpha = -1;$

repeat

$Rnd_\alpha = \text{Random}(ID_\alpha, index);$

$Found = \text{TRUE};$

for each node β in α 's two communication hops **do**

$Rnd_\beta = \text{Random}(ID_\beta, index);$

if ($Rnd_\alpha < Rnd_\beta$) or ($Rnd_\alpha == Rnd_\beta$ and $ID_\alpha < ID_\beta$);

then

$Found = \text{FALSE}; \text{break};$

end

end

if $Found$ **then**

$FreNum_\alpha = index;$

else

$index ++;$

end

until $FreNum_\alpha > -1;$

2.1.1 Exclusive Frequency Assignment. In exclusive frequency assignment, nodes first exchange their IDs among two communication hops so that each node knows its two-hop neighbors' IDs. In the second broadcast, each node beacons all neighbors' IDs it has collected during the first broadcast period.

Eavesdropping. Even though the even selection scheme leads to even sharing of available frequencies among any two-hop neighborhood, it involves a number of two-hop broadcasts. To reduce the communication cost, we propose a lightweight eavesdropping scheme.

2.2 Basic Notations

As Algorithm 1 states, for each frequency number, each node calculates a random number (Rnd_α) for itself and a random number (Rnd_β) for each of its two-hop neighbors with the same pseudorandom number generator.

Bus masters are divided into two disjoint sets, $\mathcal{M}_{RT}$ and $\mathcal{M}_{NRT}$.

RT Masters $\mathcal{M}_{RT} = \{\vec{m}_1, \dots, \vec{m}_n\}$ denotes the n RT masters issuing real-time constrained requests. To model the current request issued by an $\vec{m}_i$ in $\mathcal{M}_{RT}$, three parameters—the recurrence time (r_i), the service cycle (c_i), and the relative deadline (d_i)—are used, with their relationships.

NRT Masters $\mathcal{M}_{NRT} = \{\vec{m}_{n+1}, \dots, \vec{m}_{n+m}\}$ is a set of m masters issuing nonreal-time constrained requests. In our model, each $\vec{m}_j$ in $\mathcal{M}_{NRT}$ needs only one parameter, the service cycle, to model the current request it issues.

Here, a question may arise, since each node has a global ID. Why don't we just map nodes' IDs within two hops into a group of frequency numbers and assign those numbers to all nodes within two hops?

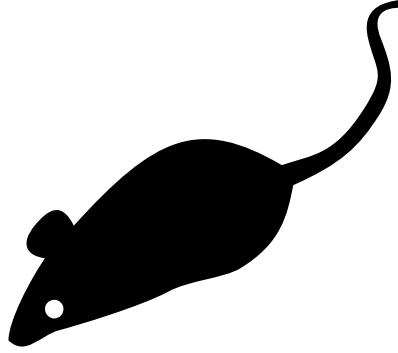


Fig. 1. Code before preprocessing.

3 Simulator

If the model checker requests successors of a state which are not created yet, the state space uses the simulator to create the successors on-the-fly. To create successor states the simulator conducts the following steps.

- (1) Load state into microcontroller model.
- (2) Determine assignments needed for resolving nondeterminism.
- (3) For each assignment.
 - (a) either call interrupt handler or simulate effect of next instruction, or
 - (b) evaluate truth values of atomic propositions.
- (4) Return resulting states.

Figure 1 shows a typical microcontroller C program that controls an automotive power window lift. The program is one of the programs used in the case study described in Section 3. At first sight, the program looks like an ANSI C program. It contains function calls, assignments, if clauses, and while loops.

3.1 Problem Formulation

The objective of variable coalescence-based offset assignment is to find both the coalescence scheme and the MWPC on the coalesced graph. We start with a few definitions and lemmas for variable coalescence.

Definition 3.1 (Coalesced Node (C-Node)). A C-node is a set of live ranges (webs) in the AG or IG that are coalesced. Nodes within the same C-node cannot interfere with each other on the IG. Before any coalescing is done, each live range is a C-node by itself.

Definition 3.2 (C-AG (Coalesced Access Graph)). The C-AG is the access graph after node coalescence, which is composed of all C-nodes and C-edges.

LEMMA 3.3. *The C-MWPC problem is NP-complete.*

PROOF. C-MWPC can be easily reduced to the MWPC problem assuming a coalescence graph without any edge or a fully connected interference graph. Therefore, each C-node is an uncoalesced live range after value separation and C-PC is equivalent to PC. A fully connected interference graph is made possible when all live ranges interfere with each other. Thus, the C-MWPC problem is NP-complete. $\square$

Table 1. Simulation Configuration

TERRAIN ^a	(200m×200m) Square
Node Number	289
Node Placement	Uniform
Application	Many-to-Many/Gossip CBR Streams
Payload Size	32 bytes
Routing Layer	GF
MAC Layer	CSMA/MMSN
Radio Layer	RADIO-ACCNOISE
Radio Bandwidth	250Kbps
Radio Range	20m–45m

Source: This is a table sourcenote. This is a table sourcenote. This is a table sourcenote.

Note: This is a table footnote.

^aThis is a table footnote. This is a table footnote. This is a table footnote.

LEMMA 3.4 (LEMMA SUBHEAD). *The solution to the C-MWPC problem is no worse than the solution to the MWPC.*

PROOF. Simply, any solution to the MWPC is also a solution to the C-MWPC. But some solutions to C-MWPC may not apply to the MWPC (if any coalescing were made). $\square$

4 Performance Evaluation

During all the experiments, the Geographic Forwarding (GF) [?] routing protocol is used. GF exploits geographic information of nodes and conducts local data-forwarding to achieve end-to-end routing. Our simulation is configured according to the settings in Table 1. Each run lasts for 2 minutes and repeated 100 times. For each data value we present in the results, we also give its 90% confidence interval.

5 Conclusions

In this article, we develop the first multifrequency MAC protocol for WSN applications in which each device adopts a single radio transceiver. The different MAC design requirements for WSNs and general wireless ad-hoc networks are compared, and a complete WSN multifrequency MAC design (MMSN) is put forth. During the MMSN design, we analyze and evaluate different choices for frequency assignments and also discuss the nonuniform back-off algorithms for the slotted media access design.

6 Typical references in new ACM Reference Format

To change from the current author-year citation style to a numeric citation style, comment out the line `\setcitestyle{authoryear}` at the top of this file and instead uncomment the line `\setcitestyle{acmnumeric}`. However, we recommend using the author-year citation style. In either citation style, to create citations that do not interrupt the sentence structure like [?] use `\cite` or `\citep`, and to use citations that include the author names as part of the sentence like ?, use `\citet`.

Here are some sample references to various types of material: a paginated journal article [?], an enumerated journal article [?], a reference to an entire issue [?], a monograph (whole book) [?], a monograph/whole book in a series (see 2a in spec. document) [?], a divisible-book such as an

anthology or compilation [?], followed by the same example, however we only output the series if the volume number is given [?] (so Editor00a's series should NOT be present since it has no vol. no.), a chapter in a divisible book [?], a chapter in a divisible book in a series [?], a multi-volume work as book [?], an article in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [?], a proceedings article with all possible elements [?], an example of an enumerated proceedings article [?], an informally published work [?], a doctoral dissertation [?], a master's thesis: [?], an online document / world wide web resource [???], a video game (Case 1) [?] and (Case 2) [?] and [?] and (Case 3) a patent [?], work accepted for publication [?], 'YYYYb'-test for prolific author [?] and [?]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [?]. Boris / Barbara Beeton: multi-volume works as books [?] and [?].

A Switching times

In this appendix, we measure the channel switching time of Micaz [?] sensor devices. In our experiments, one mote alternately switches between Channels 11 and 12. Every time after the node switches to a channel, it sends out a packet immediately and then changes to a new channel as soon as the transmission is finished. We measure the number of packets the test mote can send in 10 seconds, denoted as N_1 . In contrast, we also measure the same value of the test mote without switching channels, denoted as N_2 . We calculate the channel-switching time s as

$$s = \frac{10}{N_1} - \frac{10}{N_2}.$$

By repeating the experiments 100 times, we get the average channel-switching time of Micaz motes: $24.3 \mu\text{s}$.

B Supplementary materials

B.1 This is an example of Appendix subsection head

Channel-switching time is measured as the time length it takes for motes to successfully switch from one channel to another. This parameter impacts the maximum network throughput, because motes cannot receive or send any packet during this period of time, and it also affects the efficiency of toggle snooping in MMSN, where motes need to sense through channels rapidly.

By repeating experiments 100 times, we get the average channel-switching time of Micaz motes: $24.3 \mu\text{s}$. We then conduct the same experiments with different Micaz motes, as well as experiments with the transmitter switching from Channel 11 to other channels. In both scenarios, the channel-switching time does not have obvious changes. (In our experiments, all values are in the range of $23.6 \mu\text{s}$ to $24.9 \mu\text{s}$.)

B.2 Appendix subsection head

The primary consumer of energy in WSNs is idle listening. The key to reduce idle listening is executing low duty-cycle on nodes. Two primary approaches are considered in controlling duty-cycles in the MAC layer.